Stochastic Simulation Assignment 2
Urban Waste Collection

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1 Introduction

In this report we describe Urban Waste Collection Simulation. We present our code as well as explain and interpret our results.

2 Variables

In this report, the parameters are using the symbols as below.

Parameter	Symbol	Default value
Number of districts	N	3
Number of truck	M	3
District	i	1, 2, 3
Arrival rate	λ_i	0.4
Service rate	μ_j	1, 1.5, 1
	k_j	2, 3
Waste type probability	q_i	$\frac{1}{3}$
Waste type	j	1, 2, 3
Friendliness	p	0.5
Rerouting	K	5

TABLE 1: Default parameters

3 Assumptions

A regenerative cycle is the time of all the truck to be idle.
The stop of simulation is controlled by stopping conditions. For generative method, the simulation stop when the relative precision is reaching our setted level(≤ 0.1 in our case) or when the simulated cycles being too much and greater than our setted upper bound maximum of cycle. For the batch mean method, the simulation start at the end of warm up period and stop one batch at batch length. The total simulation stop when reached the given number of batches. When the truck reroute and return the job, the service time of it is a newly random drawn service time. All the result in this report is rounded to 3 decimals and based on random seed 42.

4 Structure

4.1 Arrival

Each truck is assigned to a home district. The requests arrive to different district as independent Poisson processes, which means the time between arrivals in each district follows an exponential distribution with λ_i and the probability of every arrival is memoryless. All the three districts start with an initial arrival event with time 0, which will build each district an independent queue.

Algorithm 1 Arrival Event

Draws a waste type, given 1, 2, or 3 from uniform U(0,1)
record in a tuple represents the request: (arrival time, waste type) and appends it to queues according to district
increment the number of total job in district i ▷ +1 on every arrival (and also will -1 on every departure)
if the home truck status is idle **then**
    call routing() function
    start service
end if
if the home truck status is service at foreign districts **then**
    if the length of queue in the district > K **then**
        schedule rerouting() ▷ at time t
    end if
end if
if the home truck status is service at home district **then**
    follow the queue and wait to do this task
end if
schedule a new arrival by interarrival time Exp(λ_i)

4.2 Service

The service consist of two parts. The first part is the truck start doing the service, the second part is the truck finish serving the task. The reason why we separate the single service to two functions is in case there exist any district get events exceed K and need to reroute truck.

Algorithm 2 Service Event(Starting)

```
assign the current task according to the queue in the current district the truck is in
record the current time and set the status of the truck to be serving
if the request supposed to be cancelled then          ▷ this boolean is determined in Rerouting()
    stop the departure procedure
elsedo service
end if
```

Algorithm 3 Service Event(finishing)

```
record the sojourn time W = t − arrivaltime with district
add the current sojourn to the cycle                  ▷ also add into the batch if is recording the batch
remove the current task from the job list and renew the queue status
change the status of the truck to idle
find next task                                        ▷ call routing()
check if all the job queue empty and truck all idle   ▷ for just finished a task it is possible to reach the cycle's end
```

4.3 Rerouting

The routing function is not a single event, but a tool function used in several events(arrival, service, and rerouting).

Algorithm 4 Routing

```
if the home queue is non-empty  then
    do service
end if
find the foreign districts in cyclic order
if foreign districts have requests to do then
    with Bernoulli probability p see if need to help
else
    stay idle
end if
```

The rerouting is triggered when an arrival event happen to check the local truck is busy in a foreign region and the task queue at home district exceeding K.

Algorithm 5 Rerouting Event

```
if truck is idle or truck is already in its home district then    ▷ check if need retouting stop rerouting
end if
if t thenhe truck need to be reroute
    cancel the current service event for this truck
    interrupt the current task and put the task to the head of the queue in the according foreign region    ▷ .insert()to
position 0
    set the status of the truck to be idle
    start service in home district        ▷ since reroute happen when home district has request, call routing()
    increment the counter of rerouting
end if
```

5 Confidence interval construction and steady state

5.1 Regenerative method

We define the regeneration points to be points in time where all trucks are idle and queues for all districts are empty.

To calculate the total waiting time we use the following:

$$W_k^{total} = \sum_{n=N_K}^{N_{K+1}-1} W_n \tag{1}$$

for $k = 1,..,N$ Where: N is the number of cycles, N_k is the number of task that arrives to an empty system at the beginning of the k-th cycle and W_n is the time it takes until finishing the n-th task.

Since we calculate this after each cycle, we get N different W^{total}. We also keep track of the number of finished tasks per cycle, which we denote M_k. At the end of the simulation we want to estimate the expected waiting time.

For each cycle we also calculate the total length of the queue as:

$$L_k^{total} = \int_{T_{N_k}}^{T_{N_{k+1}}} L(t)dt \tag{2}$$

for $k = 1,2,...,N$ Where T_{N_k} are the regeneration points and $L(t)$ is the length of the queue at the time t. Throughout the simulation we also keep track of how long each cycle took C_k. Then we estimate the average queue length with:

$$\widehat{L} = \frac{\overline{L}}{\overline{C}} = \frac{\frac{1}{N}\sum_{k=1}^{N} L_k^{\text{total}}}{\frac{1}{N}\sum_{k=1}^{N} C_k} \tag{3}$$

5.1.1 Confidence interval

We construct the approximate 95% confidence interval to calculate the expected waiting time as follows:

$$\widehat{W} \pm t_{N-1,1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\text{Var}(V)}{M^2 N}} \tag{4}$$

Where $t_{N-1,1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$ - percentile od student's t-distribution with $N-1$ degrees of freedom and $\overline{V} = \overline{W} - W\overline{M}$

We compute this value algebraically in Python.

We also construct the approximate 95% confidence interval to calculate the average queue length as follows:

$$\widehat{L} \pm t_{N-1,1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\widehat{\text{Var}}(V)}{\overline{C}^2 N}} \tag{5}$$

Where $t_{N-1,1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$ - percentile od student's t-distribution with $N-1$ degrees of freedom and $V = L^{total} - LC$

We compute this value algebraically in Python.

5.1.2 Steady-state

Thanks to computing the expectation of the waiting time, we can approximate when the simulation gets into the steady state. This is when the difference of the calculated expected waiting time and the expected waiting time in the simulation is smaller the δ. The same logic holds for the average queue length.

5.2 Batch mean method

We perform on run of the simulation and then derive the performance measures. Since at the beginning of the simulation, the warm up period, the simulation is not in a steady state, we skip this part in our calculations. To assign the length of the warm up period to be one mean cycle. After deleting this period we divide the reminder run into k = 30 batches of b= 200 observations. The warm up size D is 50 satisfying to the Rule of thumb that $b = 4D$. The value of b is not fixed, since the warm up length is reference to the mean cycle length we calculated in regenerative. We decided on this number based on [Sch82].

The chosen size can make the relative estimation precision to be small enough realizing:

$$t_{r-1,1-\frac{\alpha}{2}} \frac{\sqrt{S^2(r)}}{\sqrt{r}} \leq \frac{\delta}{1+\delta}\hat{X} \tag{6}$$

We calculate the aggregate observations $\overline{D_1},..,\overline{D_r}$ based on following formula:

$$\overline{D_1} = \frac{1}{b}\sum_{k=1}^{b} D_k, \overline{D_2} = \frac{1}{b}\sum_{k=b+1}^{2b} D_k, ..., \overline{D_r} = \frac{1}{b}\sum_{k=(r-1)b+1}^{rb} D_k \tag{7}$$

To calculate the point estimate we use the following:

$$\overline{\overline{D}}(r) = \frac{1}{r}\sum_{l=1}^{r} \overline{D_l} \tag{8}$$

5.2.1 Confidence interval

After this we can calculate the approximate $100(1-\alpha)\%$ confidence interval for W_q:

$$[\overline{\overline{D}}(r) - t_{r-1,1-\frac{\alpha}{2}} \frac{\sqrt{\overline{S}^2(r)}}{\sqrt{r}}, \overline{\overline{D}}(r) + t_{r-1,1-\frac{\alpha}{2}} \frac{\sqrt{\overline{S}^2(r)}}{\sqrt{r}}] \tag{9}$$

where $t_{r-1,1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$ - percentile od student's t-distribution with $r-1$ degrees of freedom and $\overline{S}^2(r) = \frac{1}{r-1}\sum_{k=1}^{r}(\overline{D_k} - \overline{\overline{D}}(r))^2$

We use 95% confidence interval with relative precision $\leq 10\%$.

6 Present verification and comparison results

We examine the validity by checking the two verification cases. All the confidence interval mentioned below is representing the 95% confidence intervals with relative precision ≤ 0.1 .

6.1 Verification 1

Set N = M = 1, p = 0, and use only Type 3 service, which service time is distributed by $Exp(\mu_3 = 1.0)$ with arrival rate $\lambda_1 = 0.5$. The expectation is calculated by $E[W] = \frac{1}{\mu_3 - \lambda_1}$.

Simulated waiting time		Expected waiting time
regenerative	batch mean	E[W]
1.838	2.175	2.000

TABLE 2: Verification 1(M/M/1)

The simulated waiting time of both methods are in the confidence interval.

6.2 Verification 2

Set N = M =2, p =1, $\lambda_1 = \lambda_2 = \lambda = 0.6$ equal service rates $\mu_1 = \mu_2 = \mu$, K =∞, a=1.2.
The expectation of waiting time is calculated by

$$E[W] = E[W_q] + \frac{1}{\mu} = \frac{C(2,a)}{2\mu - 2\lambda} + \frac{1}{\mu} = \frac{\frac{a^2}{2+a}}{2\mu - 2\lambda} + \frac{1}{\mu} \tag{10}$$

Simulated waiting time		Expected waiting time
regenerative	batch mean	E[W]
1.537	1.583	1.563

TABLE 3: Verification 2(M/M/2)

The simulated waiting time of both methods are in the confidence interval.

6.3 Comparison Experiment

	Simulated waiting time		Confidence interval	
	regenerative	batch mean	regenerative	batch mean
p=0.50	2.871	2.885	[2.588, 3.156]	[2.722, 3.047]
p=0.51	2.785	2.754	[2.513, 3.058]	[2.568, 2.940]

TABLE 4: Comparison experiment

From Table 4 we could see the confidence intervals have overlap with same p-value. and the simulated expected waiting time values are not the exact same but closely same to each other.Because of this overlap, we cannot conclude that increasing p by 0.01 provides a statistically significant improvement at a 95% confidence level.

6.4 Long run

With the default parameters that we mention in Table 1, we obtained the following results:

District	Regenerative method		Batch mean method	
	E[W]	95% confidence interval	E[W]	95% confidence interval
1	2.853	[2.571, 3.135]	2.822	[2.663, 2.981]
2	2.923	[2.634, 3.213]	2.833	[2.674, 2.993]
3	2.841	[2.559, 3.122]	2.930	[2.765, 3.095]
Total	2.872	[2.588, 3.156]	2.885	[2.722, 3.047]

TABLE 5: Expected waiting time per district

District	Regenerative method		Batch mean method	
	E[L]	95% confidence interval	E[L]	95% confidence interval
1	1.141	[1.028, 1.254]	1.129	[1.065, 1.192]
2	1.169	[1.054, 1.285]	1.133	[1.069, 1.197]
3	1.136	[1.024, 1.249]	1.172	[1.106, 1.238]
Total	3.446	[3.105, 3.787]	3.462	[3.267, 3.657]

TABLE 6: Expected average queue length per district

Truck	Regenerative method		Batch mean method	
	utilization	95% confidence interval	utilization	95% confidence interval
1	0.674	[0.608, 0.741]	0.666	[0.629, 0.704]
2	0.670	[0.604, 0.737]	0.669	[0.631, 0.706]
3	0.678	[0.611, 0.745]	0.662	[0.625, 0.699]

TABLE 7: Expected truck utilization per truck

The rerouting rate of regenerative method is 0, and the rerouting rate of batch mean method is 0.00035.

7 Conclusion

The result we get from regenerative and batch mean method is closely similar to each other with relative precision ≤ 0.1, which means the estimation is reliable.
The rerouting happen so less is because the average queue length is only slightly above 1, which means the friendliness trucks are most likely to return without its home district's requests exceeds K(=5 in default). And only the routing with interruption and exceed home queue is defined as rerouting.

References

[Sch82] B. W. Schmeiser. "Batch Size Effects in the Analysis of Simulation Output." In: *Operations Research* 30.556-68 (1982).

8 AI statement

No AI usage.